

3) Derivada de un polinomio de Bernoulli: $\mathcal{D} B_v^{\wedge}(x) = v B_{v-1}^{\wedge}(x)$

$$\phi(\lambda, \tau; x) = \frac{\tau^{\wedge} e^{\tau x}}{(e^{\tau} - 1)^{\wedge}} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^{\wedge}(x)$$

Derivo respecto a x :

$$\mathcal{D}_x \left[\frac{\tau^{\wedge} e^{\tau x}}{(e^{\tau} - 1)^{\wedge}} \right] = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \mathcal{D} (B_v^{\wedge}(x))$$

$$\left(\frac{\tau^{\wedge}}{(e^{\tau} - 1)^{\wedge}} e^{\tau x} \right) \tau = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \mathcal{D} (B_v^{\wedge}(x))$$

$$\tau \cdot \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^{\wedge}(x) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \mathcal{D} (B_v^{\wedge}(x))$$

$$\sum_{v=0}^{\infty} \frac{\tau^{v+1}}{(v+1)!} \cdot (v+1) B_v^{\wedge}(x) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \mathcal{D} (B_v^{\wedge}(x))$$

$$\sum_{v=1}^{\infty} \frac{\tau^v}{v!} \cdot v B_{v-1}^{\wedge}(x) = \underbrace{\frac{\tau^0}{0!} \mathcal{D} (B_0^{\wedge}(x))}_{=0} + \sum_{v=1}^{\infty} \frac{\tau^v}{v!} \mathcal{D} (B_v^{\wedge}(x))$$

$$\sum_{v=1}^{\infty} \frac{\tau^v}{v!} \cdot v B_{v-1}^{\wedge}(x) = \sum_{v=1}^{\infty} \frac{\tau^v}{v!} \mathcal{D} [B_v^{\wedge}(x)]$$

$$\text{Luego } v B_{v-1}^{\wedge}(x) = \mathcal{D} [B_v^{\wedge}(x)]$$